Lecture 5: Mild Solutions for SPDEs

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§5 Mild solutions for SPDEs

We consider (semi-linear) stochastic evolution equations on a (real separable) Hilbert space H:

$$dX_t \;=\; \big(AX_t + B(X_t)\big)\,dt \;+\; C(X_t)\,dW_t, \qquad t \in [0,T], \qquad X_0 = \xi. \tag{1}$$

Here $A\colon D(A) \subset H \to H$ is typically an *unbounded* (linear) operator, $B\colon H \to H$ is the drift nonlinearity, and $C\colon H \to L_2(U,H)$ is the diffusion coefficient (for a U-valued Q-Wiener process W).

Remark 1 (What "semi-linear" means). The equation (1) is called *semi-linear* because the (typically unbounded) operator A enters *linearly* in the highest-order part AX_t, while the remaining terms $B(X_t)$ and $C(X_t)$ are nonlinear maps of the state. In PDE language, A contains the highest-order derivatives and the nonlinearities are "lower-order" (they do not change the principal part). This contrasts with *quasi-linear* problems, where the leading operator itself depends on X (e.g. $A(X_t)X_t$).

Remark 2. If one formally integrates (1), one obtains

$$X_t \overset{?}{=} \xi + \int_0^t (AX_s + B(X_s))\,ds + \int_0^t C(X_s)\,dW_s.$$

This is *not* a good starting point in infinite dimensions because AX_s may be ill-defined unless $X_s \in D(A)$ a.s. The *mild formulation* avoids applying A directly to X.

Semigroup and mild formulation

Assume that A generates a strongly continuous semigroup $(T_t)_{t\geq 0}$ on H, i.e.

$$T_t = e^{tA}, \qquad t \geq 0, \qquad T_0 = I, \qquad T_{t+s} = T_t T_s.$$

Definition 1 (Mild solution)**.** An adapted process $(X_t)_{t\in[0,T]}$ is called a *mild solution* of (1) on $[0,T]$ if, for all $t \in [0,T]$,

$$X_t \;=\; T_t\xi + \int_0^t T_{t-s}B(X_s)\,ds + \int_0^t T_{t-s}C(X_s)\,dW_s, \tag{2}$$

where the integrals are well-defined (Bochner integral for the drift term and Itô integral for the stochastic term).

Remark 3. The stochastic term

$$\int_0^t T_{t-s}C(X_s)\,dW_s$$

is often called a *stochastic convolution*. Under standard integrability assumptions, it is an H-valued continuous square-integrable martingale.

Deterministic motivation: variation of constants

Consider the (linear) deterministic evolution equation with inhomogeneous forcing

$$\frac{d}{dt}x(t) = Ax(t) + f(t), \qquad t > 0, \qquad x(0) = x, \tag{3}$$

where $f\colon [0,T] \to H$ is a given function (independent of x). If A generates $(T_t)_{t\geq 0}$, then the corresponding variation-of-constants (Duhamel) formula is

$$x(t) \;=\; T_t x + \int_0^t T_{t-s} f(s)\, ds. \tag{4}$$

Derivation (integration factor). Formally, set $y(t) := T_{-t}x(t)$ (i.e. $y(t) = e^{-tA}x(t)$ when this makes sense). Then

$$y'(t) \;=\; \frac{d}{dt}\big(T_{-t}x(t)\big) \;=\; -T_{-t}Ax(t) + T_{-t}x'(t) \;=\; T_{-t}f(t).$$

Integrating from 0 to t gives

$$T_{-t}x(t) = x + \int_0^t T_{-s}f(s)\, ds.$$

Multiplying by T_t and using $T_tT_{-s} = T_{t-s}$ yields (4).

Homogeneous case. If $f \equiv 0$, then (3) reduces to $x'(t) = Ax(t)$ with solution $x(t) = T_t x = e^{tA}x$.

Remark 4 (Fundamental solution and convolution form). In the bounded-operator case, $T_t = e^{tA} = \sum_{k=0}^{\infty} \frac{t^k}{k!}A^k$ is the *fundamental solution* of the homogeneous equation. The inhomogeneous term in (4) is a time-convolution of f against the semigroup kernel T_{t-s}.

Remark 5. Formula (2) for the SPDE is the stochastic analogue of (4): the drift is convolved with the semigroup (now with $f(s)$ replaced by $B(X_s)$), and the noise enters through the stochastic convolution $\int_0^t T_{t-s}C(X_s)\, dW_s$.

Remarks

Remark 6 (Why mild solutions are natural). In SPDEs, A is often a differential operator (e.g. Laplacian) and $D(A)$ encodes spatial regularity. Even if $X_0 = \xi \in H$, the process X_t need not lie in $D(A)$ for all t. The mild formulation uses the smoothing effect of T_t and is the standard notion for existence/uniqueness via fixed-point arguments.

Remark 7 (C_0 property). The semigroup generated by A is assumed *strongly continuous*:

$$\lim_{t\downarrow 0} \|T_t x - x\|_H = 0 \qquad \text{for all } x \in H.$$

This is exactly what one needs for (2) to reduce to $X_0 = \xi$ at $t = 0$.

Remark 8 (Strong vs. mild). If a mild solution is sufficiently regular so that $X_t \in D(A)$ a.s. and the map $t \mapsto X_t$ is differentiable in a suitable sense, then (2) can be differentiated to recover the *strong* (differential) form (1). In general, mild solutions are weaker but typically easier to construct.

Appendix: Strongly continuous semigroups (C_0-semigroups)

Formal variation-of-constants (SPDE). In the same spirit as the deterministic integration-factor argument, one *formally* writes

$$X_t = e^{tA}\xi + \int_0^t e^{(t-s)A} B(X_s)\, ds + \int_0^t e^{(t-s)A} C(X_s)\, dW_s.$$

This is exactly the mild formula (2) when we denote $T_t = e^{tA}$.

Motivation from bounded operators. If $A \in \mathcal{L}(E)$ is bounded on a Banach space E, then

$$e^{tA} := \sum_{k=0}^{\infty} \frac{t^k}{k!} A^k$$

is well-defined for all $t \geq 0$, and it satisfies:

$$e^{0A} = I, \qquad e^{(t+s)A} = e^{tA} e^{sA},$$

and for each $u \in E$, the map $t \mapsto e^{tA}u$ is continuous on $[0, \infty)$.

Definition 2 (C_0-semigroup)**.** Let E be a Banach space. A family $(T_t)_{t\geq 0} \subset \mathcal{L}(E)$ is called a *strongly continuous semigroup* (or C_0*-semigroup*) on E if:

1. $T_0 = I_E$;
2. $T_{t+s} = T_t \circ T_s$ for all $s, t \geq 0$;
3. for every $u \in E$, the map $t \mapsto T_t u$ is continuous on $[0, \infty)$.

Definition 3 (Infinitesimal generator)**.** Let $(T_t)_{t\geq 0}$ be a C_0-semigroup on E. Its *infinitesimal generator* is the pair $(A, D(A))$ defined by

$$Au := \lim_{h \downarrow 0} \frac{T_h u - u}{h}, \qquad \text{for } u \in D(A),$$

where

$$D(A) := \left\{ u \in E : \lim_{h \downarrow 0} \frac{T_h u - u}{h} \text{ exists in } E \right\}.$$

Remark 9. The limit is *one-sided* ($h \downarrow 0$) because the semigroup is only defined for times $t \geq 0$.

Remark 10. For bounded $A \in \mathcal{L}(E)$, the semigroup $T_t = e^{tA}$ is a C_0-semigroup and its generator is exactly A with domain $D(A) = E$. In SPDEs, the important examples (Laplacian, elliptic operators, etc.) are typically unbounded, so one starts from the semigroup rather than from the exponential series.

Remark 11. For a C_0-semigroup, the generator domain $D(A)$ is dense in E (in the standard semigroup setting). This density is one reason the generator captures the semigroup uniquely.

Remark 12 (Generator is closed). If $(A, D(A))$ is the generator of a C_0-semigroup on E, then $(A, D(A))$ is a *closed* linear operator. Equivalently, its graph

$$\mathcal{G}(A) := \{(u, Au) : u \in D(A)\} \subset E \times E$$

is a closed subspace of $E \times E$. Thus, if $u_n \in D(A)$ with $u_n \to u$ in E and $Au_n \to v$ in E, then $u \in D(A)$ and $Au = v$.

Equivalently, $D(A)$ equipped with the *graph norm*

$$\|u\|_{D(A)} := \|u\|_E + \|Au\|_E$$

is complete. In concrete terms: if (u_n) is a Cauchy sequence in $D(A)$ (for the graph norm), $u_n \to u$ in E, and (Au_n) is Cauchy in E, then $u \in D(A)$ and $Au_n \to Au$ in E.

Example: Laplace operator on $E = L^2(\mathbb{R}^d)$

Let $(W_t)_{t\geq 0}$ be a standard d-dimensional Brownian motion in $\mathbb{R}^d$. For $f \in C_c^2(\mathbb{R}^d)$ and $x \in \mathbb{R}^d$, apply Itô's formula to $s \mapsto f(x + W_s)$:

$$f(x + W_t) = f(x) + \int_0^t \nabla f(x + W_s)\, dW_s + \frac{1}{2}\int_0^t \Delta f(x + W_s)\, ds. \tag{5}$$

Define

$$T_t f(x) := E\big[f(x + W_t)\big], \qquad t \geq 0.$$

Taking expectations in (5) (the stochastic integral has mean zero) yields

$$T_t f(x) - f(x) = \frac{1}{2}\int_0^t E\big[\Delta f(x + W_s)\big]\, ds = \frac{1}{2}\int_0^t T_s(\Delta f)(x)\, ds.$$

Hence

$$\frac{T_t f(x) - f(x)}{t} = \frac{1}{t}\int_0^t T_s\left(\frac{1}{2}\Delta f\right)(x)\, ds \underset{t\downarrow 0}{\longrightarrow} \frac{1}{2}\Delta f(x). \tag{6}$$

Remark 13. The family $(T_t)_{t\geq 0}$ is the *heat semigroup*. In convolution form,

$$T_t f = \Gamma_t * f,$$

where Γ_t is the heat kernel on $\mathbb{R}^d$. This immediately implies that T_t is linear and bounded on $L^2(\mathbb{R}^d)$.

Remark 14 (Semigroup property). Using independent increments of Brownian motion (or the convolution identity for heat kernels),

$$T_t(T_s f) = T_{t+s} f, \qquad s, t \geq 0.$$

Moreover, $(T_t)_{t\geq 0}$ is a C_0-semigroup on $L^2(\mathbb{R}^d)$.

Remark 15 (Generator on a core). Let $(A, D(A))$ denote the infinitesimal generator of the heat semigroup on $L^2(\mathbb{R}^d)$. Then

$$C_c^2(\mathbb{R}^d) \subset D(A), \qquad Af = \frac{1}{2}\Delta f \quad \text{for } f \in C_c^2(\mathbb{R}^d),$$

by the difference-quotient limit (6).

Remark 16 (Full domain (Sobolev description)). The generator is the closure of $\frac{1}{2}\Delta$ initially defined on $C_c^2(\mathbb{R}^d)$, and its full domain is

$$D(A) = H^{2,2}(\mathbb{R}^d) = W^{2,2}(\mathbb{R}^d),$$

with $A = \frac{1}{2}\Delta$ in the weak (distributional) sense.

Remark 17 (Weak derivatives (distributional formulation)). For $f \in L^2(\mathbb{R}^d)$, the weak derivative $\partial^\alpha f$ is characterized by

$$\langle \partial^\alpha f, g \rangle_{L^2(\mathbb{R}^d)} = (-1)^{|\alpha|} \int_{\mathbb{R}^d} f(x)\, \partial^\alpha g(x)\, dx, \qquad g \in C_c^\infty(\mathbb{R}^d),$$

whenever the left-hand side is represented by an L^2-function. This is the sense in which the Sobolev-space identity above is understood.

Example: Laplace operator on $L^2(D)$ (Dirichlet boundary condition)

Let $D \subset \mathbb{R}^d$ be an open set, and let $(W_t)_{t\geq 0}$ be a standard Brownian motion in $\mathbb{R}^d$. For $x \in D$, define the first exit time

$$\tau_D := \inf\{t \geq 0 : \; x + W_t \notin D\}.$$

Define, for $f \in L^2(D)$,

$$T_t f(x) := E\big[f(x + W_t)\, \mathbf{1}_{\{\tau_D > t\}}\big], \qquad t \geq 0, \; x \in D. \tag{7}$$

Equivalently, $(T_t)_{t\geq 0}$ is the semigroup of Brownian motion *killed* when it exits D.

Remark 18. The indicator $\mathbf{1}_{\{\tau_D > t\}}$ enforces the Dirichlet boundary condition probabilistically: mass is removed once the trajectory reaches ∂D. This is why the associated generator is the *Dirichlet Laplacian*, not the free Laplacian on $\mathbb{R}^d$.

Remark 19 (Semigroup and generator). Under standard assumptions, $(T_t)_{t\geq 0}$ defines a C_0-semigroup on $L^2(D)$. Its infinitesimal generator is the operator $A = \frac{1}{2}\Delta$ with Dirichlet boundary condition.

Remark 20 (Domain (smooth bounded domains)). If D is sufficiently regular (e.g. bounded with smooth boundary), then the generator domain can be identified as

$$D(A) = H^2(D) \cap H_0^1(D) \subset L^2(D),$$

and

$$Au = \frac{1}{2}\Delta u \qquad \text{for } u \in D(A),$$

where Δu is understood in the weak sense.

Remark 21 (Meaning of $H_0^1(D)$). The space $H_0^1(D)$ is the closure of $C_c^\infty(D)$ in $H^1(D)$. Informally, it consists of H^1-functions with zero trace on ∂D (Dirichlet condition). In one dimension, for $D = (a, b)$, this corresponds to vanishing boundary values at a and b in the trace sense.

Special case: $D = (0, 1)$ (Dirichlet) and spectral representation

Let $D = (0, 1) \subset \mathbb{R}$ and consider the Dirichlet Laplacian. The functions

$$e_k(x) := \sqrt{2}\, \sin(\pi k x), \qquad k = 1, 2, \ldots,$$

form a complete orthonormal system of $L^2(0, 1)$ and satisfy

$$\Delta e_k = \partial_{xx} e_k = -\pi^2 k^2\, e_k, \qquad e_k(0) = e_k(1) = 0.$$

For the generator $A = \frac{1}{2}\Delta$ we therefore have

$$A e_k = -\frac{\pi^2 k^2}{2}\, e_k.$$

Spectral representation of the semigroup. Let $(T_t)_{t\geq 0}$ be the C_0-semigroup generated by $A = \frac{1}{2}\Delta$ on $L^2(0,1)$ with Dirichlet boundary condition. For $f \in L^2(0,1)$ with Fourier coefficients $\langle f, e_k\rangle_{L^2(0,1)}$, one has the L^2-expansion

$$f = \sum_{k=1}^{\infty} \langle f, e_k\rangle\, e_k \qquad \text{and} \qquad T_t f = \sum_{k=1}^{\infty} e^{-\frac{\pi^2 k^2}{2} t}\, \langle f, e_k\rangle\, e_k.$$

Indeed, since $Ae_k = -\frac{\pi^2 k^2}{2} e_k$, one has

$$T_t e_k = e^{tA} e_k = e^{-\frac{\pi^2 k^2}{2} t}\, e_k.$$

Expanding $f = \sum_{k\geq 1} \langle f, e_k\rangle e_k$ and using linearity yields the displayed formula for $T_t f$. In particular, $T_t f \to 0$ in $L^2(0,1)$ as $t \to \infty$.

Remark 22 (Generator via Fourier coefficients). Formally differentiating term-by-term gives

$$\frac{d}{dt} T_t f = -\frac{\pi^2}{2} \sum_{k=1}^{\infty} k^2\, e^{-\frac{\pi^2 k^2}{2} t}\, \langle f, e_k\rangle\, e_k.$$

At $t = 0$ this suggests

$$Af = -\frac{\pi^2}{2} \sum_{k=1}^{\infty} k^2\, \langle f, e_k\rangle\, e_k,$$

with domain characterized by

$$D(A) = \left\{ f \in L^2(0,1) : \sum_{k=1}^{\infty} k^4\, |\langle f, e_k\rangle|^2 < \infty \right\},$$

which is equivalent to $D(A) = H^2(0,1) \cap H_0^1(0,1)$.

A useful sufficient condition (e.g. used in some references such as Stannat) is a decay assumption on the Fourier coefficients: if for some $\varepsilon > 0$,

$$|\langle f, e_k\rangle| = O\Big(k^{-\left(\frac{5}{2}+\varepsilon\right)}\Big), \qquad k \to \infty,$$

then

$$k^4\, |\langle f, e_k\rangle|^2 = O\big(k^{-1-2\varepsilon}\big),$$

so the defining series for $D(A)$ converges. By contrast, a bound of the form $|\langle f, e_k\rangle| = O(k^{-5/2})$ is borderline and does *not* imply $\sum_k k^4 |\langle f, e_k\rangle|^2 < \infty$.

Remark 23 (One-dimensional Sobolev regularity). In dimension $d = 1$, $H^1(0,1)$ embeds into $C([0,1])$, so every H^1-function has a (unique) continuous representative and admits well-defined boundary traces. In higher dimensions, the analogous embedding generally fails for $H^1(D)$; continuity requires Sobolev smoothness $s > \frac{d}{2}$ (in a suitable H^s scale).

Remark 24 (Convention for the factor $\frac{1}{2}$). Some references take the generator to be Δ (instead of $\frac{1}{2}\Delta$); then the exponent becomes $e^{-\pi^2 k^2 t}$. Only the time scaling changes.

Resolvent, spectrum, and the Hille–Yosida theorem

Remark 25 (Contraction estimate in the eigenbasis). In the Dirichlet one-dimensional case, the spectral representation yields

$$\|T_t f\|_{L^2(0,1)}^2 = \sum_{k=1}^{\infty} e^{-\pi^2 k^2 t} \, |\langle f, e_k \rangle|^2 \leq \sum_{k=1}^{\infty} |\langle f, e_k \rangle|^2 = \|f\|_{L^2(0,1)}^2, \qquad t \geq 0,$$

because $\{e_k\}_{k\geq 1}$ is an orthonormal basis of $L^2(0,1)$ and therefore, by Parseval,

$$\|T_t f\|_{L^2(0,1)}^2 = \sum_{k=1}^{\infty} \left| e^{-\frac{\pi^2 k^2}{2} t} \, \langle f, e_k \rangle \right|^2 = \sum_{k=1}^{\infty} e^{-\pi^2 k^2 t} \, |\langle f, e_k \rangle|^2.$$

Hence $(T_t)_{t\geq 0}$ is a contraction semigroup on $L^2(0,1)$.

Definition 4 (Resolvent set and resolvent operator)**.** Let E be a Banach space and let $(A, D(A))$ be a (closed) linear operator on E. The *resolvent set* of A is

$$\rho(A) := \{\lambda \in \mathbb{C} : \ \lambda I - A \colon D(A) \to E \text{ is bijective}\}.$$

For $\lambda \in \rho(A)$, the *resolvent operator* is

$$R(\lambda, A) := (\lambda I - A)^{-1}.$$

Remark 26 (Boundedness of the resolvent). If A is closed and $\lambda \in \rho(A)$, then $R(\lambda, A) \colon E \to D(A)$ is well-defined. Moreover, $R(\lambda, A) \in \mathcal{L}(E)$ when $D(A)$ is equipped with the graph norm, by the bounded inverse / closed graph theorem.

Theorem 1 (Closed graph theorem)**.** Let X, Y be Banach spaces and let $T \colon X \to Y$ be a linear operator with domain $\mathrm{Dom}(T) = X$. If the graph

$$\mathcal{G}(T) := \{(x, Tx) \in X \times Y : \ x \in X\}$$

is closed in $X \times Y$, then T is bounded (hence continuous), i.e. $T \in \mathcal{L}(X, Y)$.

Remark 27 (Why it applies to the resolvent). If A is closed, then $(\lambda I - A)^{-1} \colon E \to D(A)$ has closed graph when $D(A)$ is equipped with the graph norm. Since it is defined on all of E, the closed graph theorem implies it is bounded.

Definition 5 (Spectrum)**.** The *spectrum* of A is

$$\sigma(A) := \mathbb{C} \setminus \rho(A).$$

Remark 28 (Types of spectrum (informal)). One distinguishes three disjoint parts of the spectrum $\sigma(A)$ (for a closed densely defined operator $(A, D(A))$ on a Banach space E):

- *Point spectrum* $\sigma_p(A)$: $\lambda \in \sigma_p(A)$ iff $\lambda I - A$ is not injective, i.e. there exists $0 \neq f \in D(A)$ with
$$Af = \lambda f.$$
These are eigenvalues.

- *Continuous spectrum* $\sigma_c(A)$: $\lambda \in \sigma_c(A)$ iff $\lambda I - A$ is injective and has dense range, but $(\lambda I - A)^{-1}$ is not bounded as an operator defined on $\mathrm{Ran}(\lambda I - A)$ (equivalently, $\mathrm{Ran}(\lambda I - A)$ is not all of E). Intuitively: there is no eigenvector, but there are "approximate eigenvectors".

- *Residual spectrum* $\sigma_r(A)$: $\lambda \in \sigma_r(A)$ iff $\lambda I - A$ is injective but its range is not dense in E:

$$\overline{\mathrm{Ran}(\lambda I - A)} \neq E.$$

  Intuitively: solving $(\lambda I - A)u = f$ fails for "many" right-hand sides because of a genuine range defect.

For self-adjoint operators on Hilbert spaces, the residual spectrum is empty. For self-adjoint operators, the associated spectral measure can be decomposed into pure point, absolutely continuous, and singular continuous parts (Lebesgue decomposition).

Why $\sigma_r(A) = \varnothing$ for self-adjoint A. For a densely defined closed operator T on a Hilbert space one has

$$\overline{\mathrm{Ran}(T)} = (\ker T^*)^{\perp}.$$

Taking $T = \lambda I - A$ and using $A = A^*$ (hence $(\lambda I - A)^* = \lambda I - A$ for $\lambda \in \mathbb{R}$), injectivity of $\lambda I - A$ implies $\ker(\lambda I - A) = \{0\}$ and therefore

$$\overline{\mathrm{Ran}(\lambda I - A)} = (\ker(\lambda I - A)^*)^{\perp} = H,$$

so the range is automatically dense and λ cannot lie in the residual spectrum.

Lebesgue decomposition of the spectral measure. By the spectral theorem there is a projection-valued measure $E : \mathcal{B}(\mathbb{R}) \to \mathcal{L}(H)$ (called the *spectral measure* or *spectral resolution* of A). For each Borel set $B \subset \mathbb{R}$, the operator $E(B)$ is the corresponding *spectral projection* (an orthogonal projection onto the spectral subspace of A associated with B). Thus

$$A = \int_{\mathbb{R}} \lambda \, dE(\lambda).$$

For each $u \in H$, the scalar measure $\mu_u(B) := \langle E(B)u, u\rangle$ (the *scalar spectral measure* of A associated with u) admits a unique decomposition

$$\mu_u = \mu_u^{pp} + \mu_u^{ac} + \mu_u^{sc},$$

where, informally,

- μ_u^{pp} is *pure point*: it is a sum of atoms (Dirac masses),

$$\mu_u^{pp} = \sum_j a_j \, \delta_{\lambda_j}, \qquad a_j > 0,$$

  corresponding to the eigenvalue part of the spectrum (spectral mass sitting at isolated points).

- μ_u^{ac} is *absolutely continuous* w.r.t. Lebesgue measure: $\mu_u^{ac} \ll d\lambda$, hence $\mu_u^{ac} = w(\lambda)\, d\lambda$ for some density $w \in L^1(\mathbb{R})$. This corresponds to a "continuum" of spectral values (no atoms).

- μ_u^{sc} is *singular continuous*: it has no atoms (so no point masses), but is singular w.r.t. Lebesgue measure,

$$\mu_u^{sc} \perp d\lambda,$$

  meaning it is supported on a Lebesgue null set while still varying continuously (often with fractal-type support).

Remark 29 (Orthogonal projections and projection-valued measures). An *orthogonal projection* on a Hilbert space H is a bounded linear operator $P \in \mathcal{L}(H)$ satisfying

$$P^2 = P, \qquad P^* = P.$$

Equivalently, P projects onto the closed subspace $M = \mathrm{Ran}(P)$ in the geometric sense that every $u \in H$ decomposes uniquely as

$$u = Pu + (u - Pu),$$

where $Pu \in M$ and the "error" $u - Pu$ is orthogonal to M, i.e. $u - Pu \in M^\perp$ (so $\langle u - Pu, m\rangle = 0$ for all $m \in M$).

A *projection-valued measure* (PVM) is a map $E : \mathcal{B}(\mathbb{R}) \to \mathcal{L}(H)$ such that $E(B)$ is an orthogonal projection for every Borel set B, $E(\emptyset) = 0$, $E(\mathbb{R}) = I$, and for pairwise disjoint sets B_n,

$$E\left(\bigcup_{n=1}^{\infty} B_n\right) u = \sum_{n=1}^{\infty} E(B_n)u, \qquad u \in H,$$

(countable additivity in the strong sense). One also has $E(B \cap C) = E(B)E(C)$.

Remark 30 (Intuition for the spectral theorem). In finite dimensions, a self-adjoint matrix admits an orthonormal eigenbasis and can be diagonalized as

$$A = U \operatorname{diag}(\lambda_1, \dots, \lambda_n)\, U^*,$$

equivalently $A = \sum_{k=1}^{n} \lambda_k P_k$ with orthogonal projections P_k onto the eigenspaces. The spectral theorem generalizes this: when the spectrum is not purely discrete, the sum over eigenvalues is replaced by an operator-valued integral, with the projection-valued measure $E(\cdot)$ playing the role of "spectral projections" onto the part of the space where the spectral values lie in a set B.

Theorem 2 (Spectral theorem (self-adjoint case))**.** Let H be a Hilbert space and let A be a self-adjoint operator on H. Then there exists a unique projection-valued measure $E(\cdot)$ on $\mathbb{R}$ such that

$$A = \int_{\mathbb{R}} \lambda \, dE(\lambda),$$

and, more generally, for every bounded Borel function $f : \mathbb{R} \to \mathbb{C}$ one can define a bounded operator

$$f(A) := \int_{\mathbb{R}} f(\lambda)\, dE(\lambda).$$

Moreover, for $u, v \in H$ the map $B \mapsto \langle E(B)u, v\rangle$ is a (complex) measure, and for $u \in H$ the scalar measure $\mu_u(B) := \langle E(B)u, u\rangle$ describes how u decomposes across the spectrum of A.

Theorem 3 (Hille–Yosida (one standard form))**.** Let E be a Banach space and let $(A, D(A))$ be a linear operator. Fix constants $M \geq 1$ and $\omega \in \mathbb{R}$. Then the following are equivalent:

1. A generates a C_0-semigroup $(T_t)_{t\geq 0}$ on E with growth bound

$$\|T_t\|_{\mathcal{L}(E)} \leq M e^{\omega t}, \qquad t \geq 0.$$

2. A is closed and densely defined, and for every $\lambda > \omega$ one has $\lambda \in \rho(A)$ and for every $n \in \mathbb{N}$,

$$\|R(\lambda, A)^n\|_{\mathcal{L}(E)} \leq \frac{M}{(\lambda - \omega)^n}.$$

Remark 31 (Intuition for Hille–Yosida). The Hille–Yosida conditions say that "$\lambda I - A$ can be inverted to the right of ω" and that these inverses have the same scaling as the resolvent of a generator. Heuristically, for $\lambda > \omega$ one expects the Laplace transform identity

$$R(\lambda, A) = \int_0^\infty e^{-\lambda t} T_t \, dt,$$

which immediately gives the bound for $n = 1$,

$$\|R(\lambda, A)\| \lesssim \frac{M}{\lambda - \omega},$$

The same idea yields the higher-power estimates: for $n \geq 2$ one has the identity

$$R(\lambda, A)^n = \frac{1}{(n-1)!} \int_0^\infty t^{n-1} e^{-\lambda t} T_t \, dt,$$

hence

$$\|R(\lambda, A)^n\| \lesssim \frac{M}{(\lambda - \omega)^n}.$$

Conversely, Hille–Yosida says that these resolvent bounds are not only necessary but (together with closedness and density of $D(A)$) also sufficient to reconstruct a C_0-semigroup.

Remark 32 (Resolvent as Laplace transform of the semigroup). If $(T_t)_{t \geq 0}$ is a C_0-semigroup with $\|T_t\| \leq M e^{\omega t}$ and $\lambda > \omega$, then

$$R(\lambda, A) f = \int_0^\infty e^{-\lambda t} \, T_t f \, dt, \qquad f \in E,$$

where the integral converges absolutely in E. This identity can be checked by differentiating $t \mapsto T_t f$ for $f \in D(A)$ (so that $T_t f \in D(A)$ and $\frac{d}{dt} T_t f = A T_t f$) and integrating by parts:

$$A\left(\int_0^\infty e^{-\lambda t} T_t f \, dt\right) = \int_0^\infty e^{-\lambda t} A T_t f \, dt = \int_0^\infty e^{-\lambda t} \frac{d}{dt}(T_t f) \, dt = -f + \lambda \int_0^\infty e^{-\lambda t} T_t f \, dt.$$

Equivalently, $(\lambda I - A) R(\lambda, A) f = f$.